Lectures 13/14

Math 4741: Mathematical Analysis

Chapter 7: Queueing Theory

Texts: 1) Markov Processes for Stochastic Modeling – Oliver Ibe (Chapter 7)

2) Notes

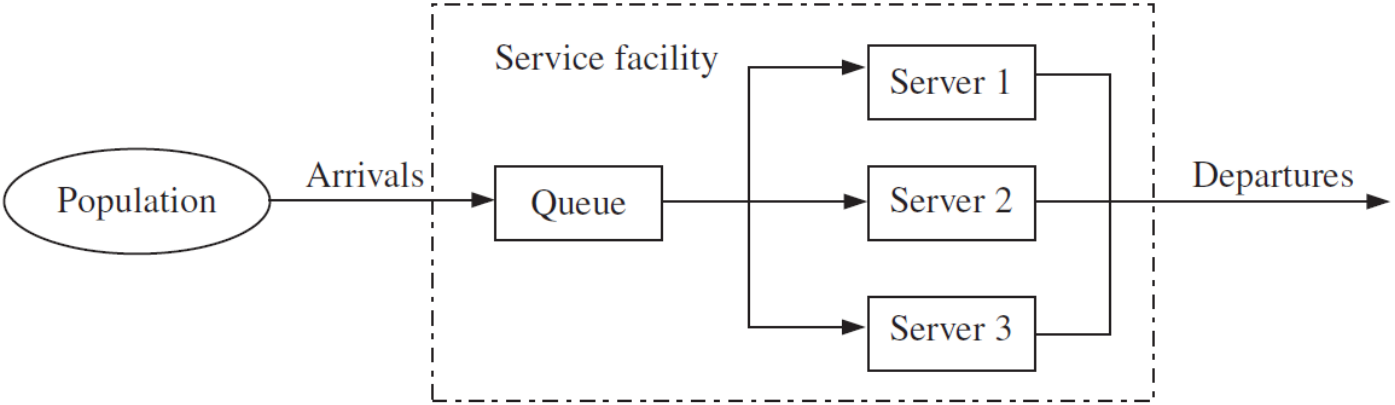
Introduction:

A queue is a waiting line of customers/jobs/items

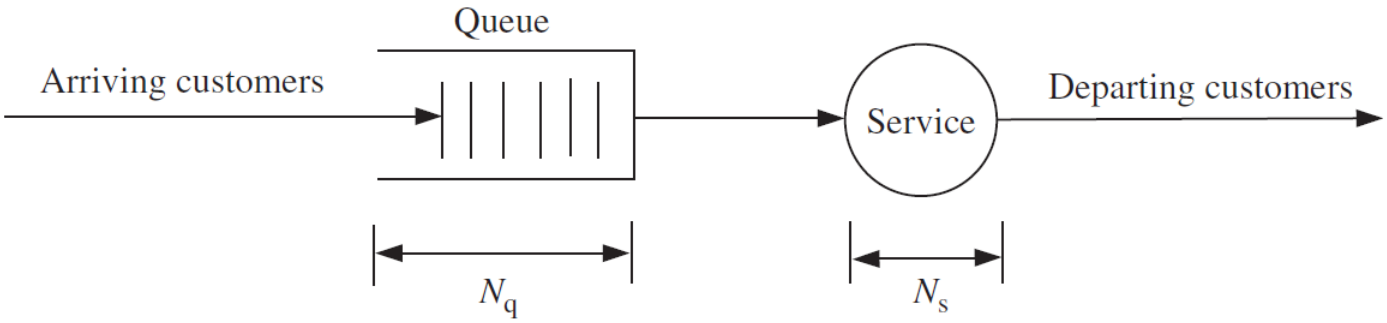
- Arise in many of our daily activities
- We join a queue to
    - Buy stamps at a post office
    - Cash cheques or deposit money at the bank
    - Pay for groceries at a super shop
    - Buy fast food or take deliveries at KFC
    - Arriving packets join a queue in a router, known as buffer

- Discuss a class of queueing system, known as Markovian queueing systems
    - Either the service time distributions are exponential, or
    - The arrival process is Poisson
        - Interarrival times are exponential
    - Or both
- Our focus will be on the steady-state analysis of the Markovian Queueing systems

Components of a Queueing system



Queueing Process



Notations

a. N_q is the number of customers in queue.
b. N_s is the number of customers receiving service.
c. N is the total number of customers in the system: $N = N_q + N_s$.
d. W is the time a customer spends in queue before going to service, i.e., the waiting time.

e. X is the time a customer spends in actual service.
f. T is the total time a customer spends in the system (also called the sojourn time): $T = W + X$.

1. *Population*, which is the source of the customers arriving at the service facility. The population can be finite or infinite.
2. *Arriving pattern*, which defines the customer interarrival process.
3. *Service time distribution*, which defines the time taken to serve each customer.
4. *Capacity of the queueing facility*, which can be finite or infinite. If the capacity is finite, customers that arrive when the system is full are lost (or blocked). Thus, a finite-capacity system is a blocking system.
5. *Number of servers*, which can be one or more than one. A queueing system with one server is called a single-server system, otherwise it is called a multiserver system. A
6. *Queueing discipline*, which is also called the *service discipline*. It defines the rule that governs how the next customer to receive service is selected after a customer who is

FCFS, LCFS, SIRO, Priority

Kendal Notation

The Kendall notation is a shorthand notation that is used to describe queueing systems. It is written in the form:

$$A/B/c/D/E/F$$

- "A" describes the arrival process (or the interarrival time distribution), which can be an exponential or nonexponential (i.e., general) distribution.
- "B" describes the service time distribution.
- "c" describes the number of servers at the queueing station.
- "D" describes the system capacity, which is the maximum number of customers allowed in the system including those currently receiving service; the default value is infinity.
- "E" describes the size of the population from where arrivals are drawn; the default value is infinity.
- "F" describes the queueing (or service) discipline. The default is FCFS.

When default values of D, E, and F are used, we use the notation A/B/c, which means a queueing system with infinite capacity, customers arrive from an infinite population, and service in an FCFS manner.

Symbols traditionally used for A and B are

- GI, which stands for general independent interarrival time.
- G, which stands for general service time distribution.
- M, which stands for memoryless (or exponential) interarrival time or service time distribution. Note that an exponentially distributed interarrival time means that customers arrive according to a Poisson process.
- D, which stands for deterministic (or constant) interarrival time or service time distribution.

Common Queueing systems

- M/M/1 queue, which is a queueing system with exponentially distributed interarrival time, exponentially distributed service time, a single server, infinite capacity, customers are drawn from an infinite population, and service is on an FCFS basis.
- M/D/1 queue, which is a queueing system with exponentially distributed interarrival time, constant service time, a single server, infinite capacity, customers are drawn from an infinite population, and service is on an FCFS basis.
- M/G/3/20 queue, which is a queueing system with exponentially distributed interarrival time, general (or nonexponentially) distributed service time, three servers, a finite capacity of 20 (i.e., a maximum of 20 customers can be in the system, including the three that can be in service at the same time), customers are drawn from an infinite population, and service is on an FCFS basis.

Little's Formula

Establish a very simple relationship among

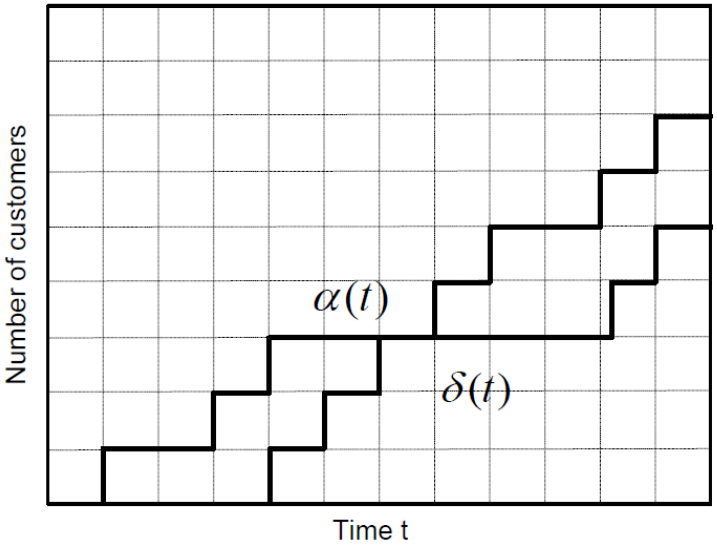
- Mean number in the system
- Mean arrival rate
- Mean system time

$$\alpha(t) = \text{number of arrivals in } (0, t)$$

$$\delta(t) = \text{number of departures in } (0, t)$$

Number of customers in the system at time t

$$N(t) = \alpha(t) - \delta(t)$$



Area between two curves

- Total time of all customers have spent during $(0, t)$
- Denote this area by $\gamma(t)$

If $\lambda(t)$ is the average arrival rate during $(0, t)$

$$\lambda_t = \frac{\alpha(t)}{t}$$

If T_t is the system time per customer

$$T_t = \frac{\gamma(t)}{\alpha(t)}$$

If $\overline{N_t}$ is the average number of customers in the queueing system during $(0, t)$

$$\overline{N}_t = \frac{\gamma(t)}{t}$$

From the last 3 equations

$$\overline{N}_t = \lambda_t T_t$$

Let us now assume that our queueing system is such that the following limits exist as $t \to \infty$.

$$\lambda = \lim_{t \to \infty} \lambda_t$$

$$T = \lim_{t \to \infty} T_t$$

Then, $\overline{N} = \lambda T$

This results is known as Little's result. It states that the average number of customers in a queueing system is equal to the average arrival rate of customers to that system, times the average time spent in that system.

M/M/1 Queueing System

- Customers arrive according to a Poisson process
  - To a single server facility
- Time to server each customer is exponential
- Default: infinite capacity, infinite population and FCFS

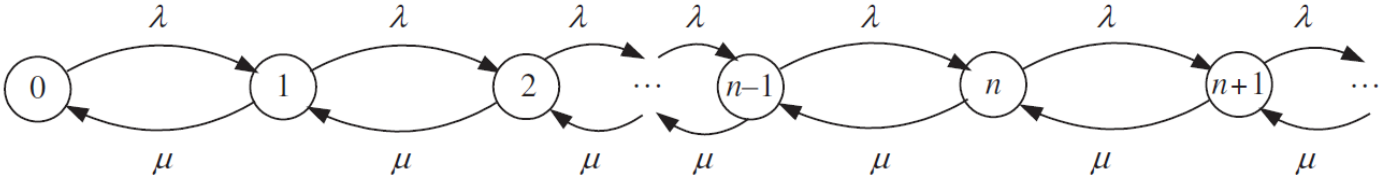


Figure 7.3 State transition rate diagram for M/M/1 queue.

$$P_n = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1\left(1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}\right)}, \quad n \geq 1$$

$$P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}}$$

$$P_n = P_0\left(\frac{\lambda}{\mu}\right)^n$$

$$P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\left(\frac{\lambda}{\mu}\right)^n}$$

$$P_0 = 1 - \frac{\lambda}{\mu}$$

Let p_n be the limiting-state probability that the process is in state n, $n = 0, 1, 2, \ldots$. Then applying the balance equations, we obtain:

$$\lambda p_0 = \mu p_1 \Rightarrow p_1 = \left(\frac{\lambda}{\mu}\right) p_0 = \rho p_0$$

$$(\lambda + \mu) p_1 = \lambda p_0 + \mu p_2 \Rightarrow p_2 = \rho p_1 = \rho^2 p_0$$

$$(\lambda + \mu) p_2 = \lambda p_2 + \mu p_3 \Rightarrow p_3 = \rho p_2 = \rho^3 p_0$$

where $\rho = \lambda/\mu$. Similarly, it can be shown that

$$p_n = \rho^n p_0 \quad n = 0, 1, 2, \ldots$$

Because

$$\sum_{n=0}^{\infty} p_n = 1 = p_0 \sum_{n=0}^{\infty} \rho^n = \frac{p_0}{1-\rho} \quad \text{where} \quad \rho < 1$$

we obtain

$$p_0 = 1 - \rho$$
$$p_n = (1-\rho)\rho^n \quad n = 0, 1, 2, \ldots; \rho < 1$$

Since $p_0 = 1 - \rho \Rightarrow \rho = 1 - p_0$ is the probability that the system is not empty

- Server is not empty
- ρ called the server utilization (utilization factor)

The expected number of customers in the system is given by

$$E[N] = \sum_{n=0}^{\infty} np_n = \sum_{n=0}^{\infty} n(1-\rho)\rho^n = (1-\rho)\sum_{n=0}^{\infty} n\rho^n$$

But

$$\frac{\mathrm{d}}{\mathrm{d}\rho}\sum_{n=0}^{\infty}\rho^n = \sum_{n=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}\rho}\rho^n = \sum_{n=1}^{\infty} n\rho^{n-1} = \frac{1}{\rho}\sum_{n=0}^{\infty} n\rho^n$$

Thus,

$$\sum_{n=0}^{\infty} n\rho^n = \rho\frac{\mathrm{d}}{\mathrm{d}\rho}\sum_{n=0}^{\infty}\rho^n = \rho\frac{\mathrm{d}}{\mathrm{d}\rho}\left(\frac{1}{1-\rho}\right) = \frac{\rho}{(1-\rho)^2}$$

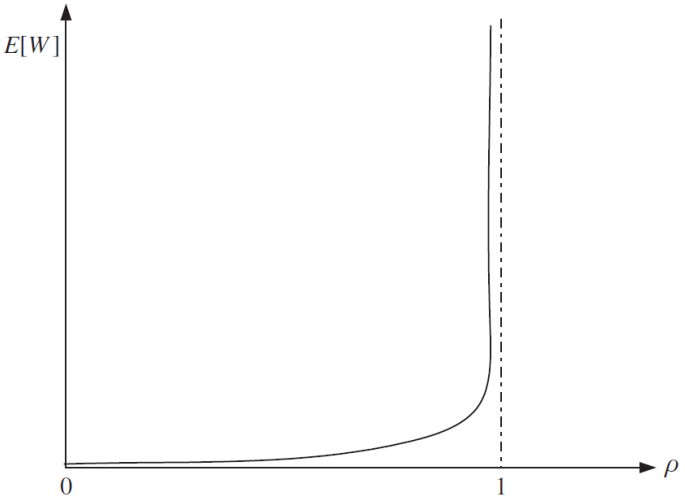
$$E[N] = (1-\rho)\sum_{n=0}^{\infty} n\rho^n = (1-\rho)\left\{\frac{\rho}{(1-\rho)^2}\right\} = \frac{\rho}{1-\rho}$$

Applying the Little's formula

$$E[T] = \frac{E[N]}{\lambda} = \frac{\lambda/\mu}{\lambda(1-\rho)} = \frac{1}{\mu(1-\rho)} = \frac{E[X]}{1-\rho}$$

$$E[W] = E[T] - E[X] = \frac{E[X]}{1-\rho} - E[X] = \frac{\rho E[X]}{1-\rho} = \frac{\rho}{\mu(1-\rho)}$$

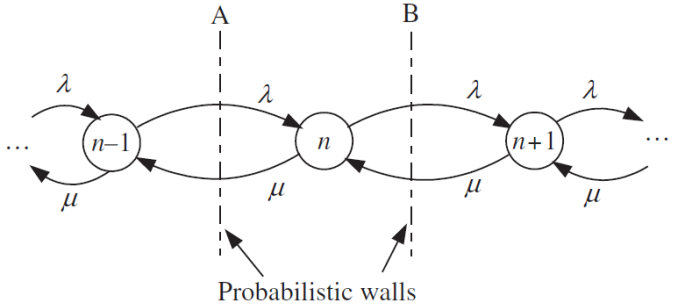
$$E\left[N_q\right] = \lambda E[W] = \frac{\lambda\rho}{\mu(1-\rho)} = \frac{\rho^2}{1-\rho}$$



Mean number of customers in service $E[N_s]$

$$E[N_s] = E[N] - E[N_q] = \frac{\rho}{1-\rho} - \frac{\rho^2}{1-\rho} = \rho$$

Stochastic Balance



Rate at which moves from left to right across a probabilistic wall is equal to the rate at which moves from right to left

For wall A, $\lambda p_{n-1} = \mu p_n$

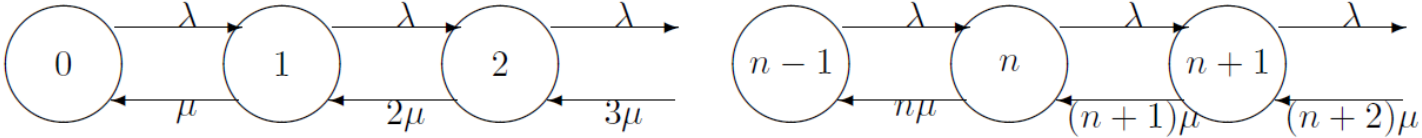
M/M/∞

In this system, we have

$$\lambda_n = \lambda, \quad n = 0, 1, 2, \cdots$$

$$\mu_n = n\mu, \quad n = 1, 2, \cdots$$

The state transition diagram is shown below:



$$P_n = P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu} = P_0 \frac{(\lambda/\mu)^n}{n!}$$

From the general solution (5.2) of the birth-death process, we have

$$P_n = P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu} = P_0 \frac{(\lambda/\mu)^n}{n!}$$

where

$$\begin{aligned} P_0 &= 1 \Big/ \left(1 + \sum_{n=1}^{\infty} \frac{\lambda^n}{\mu^n n!} \right) \\ &= 1 \Big/ \left(\sum_{n=0}^{\infty} \frac{(\lambda/\mu)^n}{n!} \right) \\ &= 1/e^{\lambda/\mu} \end{aligned}$$

Thus,

$$P_n = \frac{(\lambda/\mu)^n}{n!} e^{-\lambda/\mu}, \quad n = 0, 1, 2, \cdots$$

$N \sim Poisson\ (\lambda/\mu)$

From this (having the form of Poisson distribution), we have

$$\overline{N} = \frac{\lambda}{\mu}$$

Little's result provides

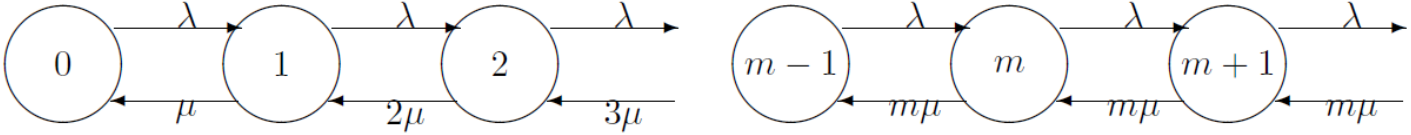
$$T = \frac{1}{\mu}$$

$M/M/m$ or $M/M/c$ Queueing Systems

We consider a system with unlimited waiting room and with a constant arrival rate λ. The system provides for a maximum m servers. Since this is still a birth-death process, we have

$$\begin{aligned}
\lambda_n &= \lambda, \quad n = 0, 1, 2, \cdots \\
\mu_n &= \min(n\mu, m\mu) \\
&= \begin{cases} n\mu, & 0 \le n < m \\ m\mu, & n \ge m \end{cases}
\end{aligned}$$

The state transition diagram is shown below:



For $n < m$,

$$\begin{aligned} P_n &= P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu} \\ &= P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!} \end{aligned}$$

For $n \geq m$,

$$\begin{aligned} P_n &= P_0 \prod_{i=0}^{m-1} \frac{\lambda}{(i+1)\mu} \prod_{j=m}^{n-1} \frac{\lambda}{m\mu} \\ &= P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{m! m^{n-m}} \end{aligned}$$

$$\sum_{n=0}^{\infty} p_n = 1 \Rightarrow p_0 = \frac{1}{\sum_{n=0}^{c-1} \left(\frac{\lambda}{\mu}\right)^n \left(\frac{1}{n!}\right) + \left(\frac{\lambda}{\mu}\right)^c \left(\frac{1}{c!}\right)\left(\frac{c\mu}{c\mu - \lambda}\right)}$$

$$\sum_{k=c}^{\infty} \left(\frac{\lambda}{\mu}\right)^c \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^{k-c} \frac{1}{c^{k-c}} = \left(\frac{\lambda}{\mu}\right)^c \frac{1}{c!} \sum_{k=c}^{\infty} \left(\frac{\lambda}{\mu}\right)^{k-c} \frac{1}{c^{k-c}} = \left(\frac{\lambda}{\mu}\right)^c \frac{1}{c!} \sum_{k=c}^{\infty} \left(\frac{\lambda}{c\mu}\right)^{k-c}$$

$$= \left(\frac{\lambda}{\mu}\right)^c \frac{1}{c!} \sum_{k=0}^{\infty} \left(\frac{\lambda}{c\mu}\right)^k = \left(\frac{\lambda}{\mu}\right)^c \frac{1}{c!} \frac{1}{1-\lambda/c\mu} = \left(\frac{\lambda}{\mu}\right)^c \frac{1}{c!} \frac{c\mu}{c\mu-\lambda}$$

The probability that an arriving customer has to wait, referred to as the delay probability, is

$$P_W = P[N \geq c] = P_c + P_{c+1} + P_{c+2} + \cdots$$

$$= \frac{P_c}{1-\frac{\lambda}{c\mu}} = \frac{P_c}{1-\rho}$$

The mean queue length is

$$E[N_q] = \sum_{n=c}^{\infty} np_n = p_c \sum_{n=0}^{\infty} n\rho^n = \frac{\rho p_c}{(1-\rho)^2} = \frac{\rho P_W}{1-\rho}$$

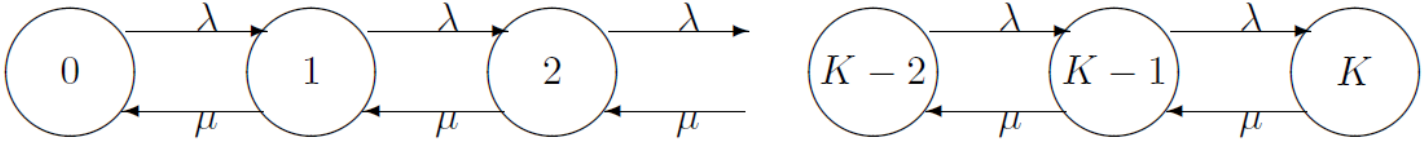
Using Little's formula, the mean waiting time

$$E[W] = \frac{E[N_q]}{\lambda} = \frac{p_c}{c\mu(1-\rho)^2} = \frac{P_W}{c\mu(1-\rho)}$$

$M/M/1/K$ Queueing Systems

We now consider a queueing system in which there is a maximum number of customers that may be stored; in particular, we assume the system can hold at most a total of K customers and any further arriving customers will be in fact be refused entry to the system and will depart immediately without service.

The state transition diagram is shown below:



$$\lambda_n = \begin{cases} \lambda, & n < K \\ 0, & n \geq K \end{cases}$$

$$\mu_n = \mu, \quad n = 1, 2, \cdots, K$$

$$
\begin{aligned}
P_n &= P_0 \prod_{i=0}^{n-1} \frac{\lambda}{\mu} \\
&= P_0 \left(\frac{\lambda}{\mu}\right)^n, \quad n \leq K
\end{aligned}
$$

$$P_n = 0, \quad n > K$$

$$
\begin{aligned}
P_0 &= \left(1 + \sum_{k=1}^{K} \left(\frac{\lambda}{\mu}\right)^k\right)^{-1} \\
&= \left(1 + \frac{(\lambda/\mu)(1-(\lambda/\mu)^K)}{1-\lambda/\mu}\right)^{-1} \\
&= \frac{1-\lambda/\mu}{1-(\lambda/\mu)^{K+1}}
\end{aligned}
$$

$$
P_n = \begin{cases} \frac{1-\lambda/\mu}{1-(\lambda/\mu)^{K+1}} \left(\frac{\lambda}{\mu}\right)^n, & 0 \leq n \leq K \\ 0, & \text{Otherwise} \end{cases}
$$

$$E[N] = \sum_{n=0}^{K} np_n = p_0 \sum_{n=0}^{K} n\rho^n = \rho p_0 \frac{\mathrm{d}}{\mathrm{d}\rho} \left\{ \frac{1-\rho^{K+1}}{1-\rho} \right\}$$

$$= \rho p_0 \left\{ \frac{1-\rho^{K+1}-(1-\rho)(K+1)\rho^K}{(1-\rho)^2} \right\} = \rho p_0 \left\{ \frac{1-(K+1)\rho^K+K\rho^{K+1})}{(1-\rho)^2} \right\}$$

$$= \frac{\rho}{1-\rho^{K+1}} \left\{ \frac{1-(K+1)\rho^K+K\rho^{K+1})}{1-\rho} \right\} = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}}$$

$$E\left[N_q\right] = E[N] - (1-p_0) = E[N] - \frac{\rho(1-\rho^K)}{1-\rho^{K+1}} = \frac{\rho}{1-\rho} - \frac{\rho(1+K\rho^K)}{1-\rho^{K+1}}$$

Customers are turned away with probability

$$p_K = \frac{(1-\rho)\rho^K}{1-\rho^{K+1}}$$

The actual rate at which customers arrive into the system, λ_A, as

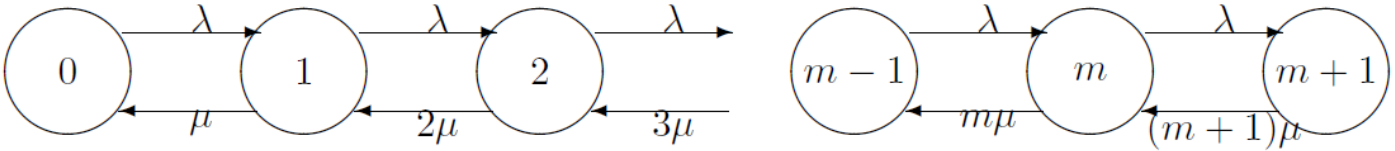
$$\lambda_A = \lambda(1-p_K) = \frac{\lambda(1-\rho^K)}{1-\rho^{K+1}}$$

We can then apply Little's formula to obtain

$$E[T] = \frac{E[N]}{\lambda_A} = \frac{1}{\mu}\left\{\frac{1}{1-\rho} - \frac{K\rho^K}{1-\rho^K}\right\}$$

$$E[W] = \frac{E[N_q]}{\lambda_A} = \frac{1}{\mu}\left\{\frac{\rho}{1-\rho} - \frac{K\rho^K}{1-\rho^K}\right\}$$

$M/M/m/m$ Queueing Systems



$$\lambda_n = \begin{cases} \lambda, & n < m \\ 0, & n \geq m \end{cases}$$

$$\mu_n = n\mu, \quad n = 1, 2, \cdots, m$$

$$P_n = P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu}, \quad n \leq m$$

$$P_n = \begin{cases} P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!}, & n \leq m \\ 0, & n > m \end{cases}$$

$$P_0 = \left(\sum_{k=0}^{m} \left(\frac{\lambda}{\mu}\right)^k \frac{1}{k!} \right)^{-1}$$

$$P_m = \frac{(\lambda/\mu)^m/m!}{\sum_{k=0}^{m}(\lambda/\mu)^k/k!}$$

This equation is also referred to as Erlang's B formula.

Or Erlang's loss formula

The actual average arrival rate is $\lambda_A = \lambda(1 - P_c)$

Because no customer is allowed to wait, $E[W]$ and $E[N_q]$ are both zero.

Mean number of customers in the system is

$$E[N] = \sum_{n=0}^{c} np_n = p_0 \sum_{n=1}^{c} n \frac{(\lambda/\mu)^n}{n!} = (\lambda/\mu) p_0 \sum_{n=1}^{c} \frac{(\lambda/\mu)^{n-1}}{(n-1)!}$$

$$= (\lambda/\mu) p_0 \sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} = (\lambda/\mu)[1 - p_c]$$

By Little's formula,

$$E[T] = E[N]/\lambda_A = 1/\mu$$

Ex 5.3.2 Consider a cellular system where handoff calls are prioritized over originating calls by assigning a portion of channels exclusively for handoff calls. Let's assume that S_R channels are assigned exclusively for handoff calls among S channels in a cell. The remaining $S_C(= S - S_R)$ channels are shard by both calls. Find the blocking probability of originating call and handoff call. Assume call arrival and departure follows Poisson processes. And the arrival rates of originating and handoff calls are λ_o and λ_h, respectively, and the service rate is μ.

$$n\mu P_n = (\lambda_0 + \lambda_h) P_{n-1} \quad 0 \le n \le S_c$$

$$n\mu P_n = \lambda_h P_{n-1} \quad S_c < n \le S$$

$$P_n = \begin{cases} \frac{(\lambda_0+\lambda_h)^n}{\mu^n n!} P_0, & 0 \le n \le S_c \\ \frac{(\lambda_0+\lambda_h)^{S_c} \lambda_h^{n-S_c}}{\mu^n n!} P_0, & S_c < n \le S \end{cases}$$

$$P_0 = \left(\sum_{i=0}^{S_c} \frac{(\lambda_0 + \lambda_h)^i}{\mu^i i!} + \sum_{i=S_c+1}^{S} \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{i-S_c}}{\mu^i i!} \right)^{-1}$$

Let P_b and P_d denote blocking probability and handoff dropping probability.

$$P_b = \sum_{n=S_c}^{S} P_n$$

$$P_d = P_S = \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{S-S_c}}{\mu^S S!} P_0$$